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ECONOMICS

Constant Utility — The Quiet Fight Over How Inflation Is Measured

Every indexed pension, tax bracket and inflation-linked bond hangs off a quantity that cannot be observed, only defined — and the definition has been fought over, from both directions, for a century.

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In the spring of 1944 the American labour movement went to war with a statistic. Wartime wage increases had been capped by the National War Labor Board at fifteen per cent above January 1941 levels — the Little Steel formula — and the cap was enforced against the Bureau of Labor Statistics' cost-of-living index, which by January 1944 had risen 23.5 per cent. Union economists produced their own accounting and concluded that the true increase was 43.5 per cent, very nearly double. Two committees of eminent economists were convened to adjudicate; the second, chaired by Wesley Mitchell of the National Bureau of Economic Research, found that the Bureau had done a competent job and that its index had understated the rise by at most three or four per cent. George Meany of the American Federation of Labor put the union objection in a sentence that has outlived everyone involved: if the index is a retail price index and not actually a cost-of-living index, he told the committee, "we have no particular interest in what the index has done in the past" [1].

Fifty-two years later the same index was on trial again, and the charge had reversed. In December 1996 five economists appointed by the Senate Finance Committee reported that the Consumer Price Index overstated increases in the cost of living, by about 1.1 percentage points a year [2]. Same instrument, opposite complaint. What had changed was not the arithmetic. It was who was paying.

The stakes are not abstract. Nearly seventy-one million Social Security beneficiaries received a 2.8 per cent increase in January 2026 because one variant of the index rose by that much between the third quarter of 2024 and the third quarter of 2025 [3]. Tax brackets, the standard deduction, federal poverty thresholds, the coupons on inflation-linked Treasury debt and a long tail of private contracts all move when the number moves. And the quantity all of this machinery claims to track cannot be observed at all — not with a better survey, not with more money, not in principle.

The Question Behind the Number

A price index looks like an average and is really the answer to a counterfactual. The economic question is not "what happened to prices" but "how much more money would a household need, at this year's prices, to be exactly as well off as it was last year." The Russian economist A. A. Konüs formalised that

question in the 1920s, in a paper that reached English-language readers through *Econometrica* in 1939 [4]. Write $e(\mathbf{p}, \mathbf{u})$ for the expenditure function: the smallest outlay that buys utility level $\mathbf{u}$ when prices are $\mathbf{p}$. Then the true cost-of-living index between two price vectors, at reference utility $\mathbf{u}$, is a ratio of two minimum outlays:

$$K(\mathbf{p}^0, \mathbf{p}^1; \mathbf{u}) = \frac{e(\mathbf{p}^1, \mathbf{u})}{e(\mathbf{p}^0, \mathbf{u})}$$

Every term on the right solves an optimisation problem that nobody observes. Utility is not measurable, and neither is the household's minimising choice at prices it never faced. So statistical agencies do the one thing they can, which is to price an actual basket of actual goods. Fix the base-period quantities $\mathbf{q}^0$ and you get the Laspeyres index; fix the current-period quantities $\mathbf{q}^1$ and you get the Paasche index:

$$P_L = \frac{\sum_i p_i^1 q_i^0}{\sum_i p_i^0 q_i^0} \quad P_P = \frac{\sum_i p_i^1 q_i^1}{\sum_i p_i^0 q_i^1}$$

Neither equals K , but the direction of the error is known rather than guessed. Under the standard assumption that households minimise the cost of reaching their chosen standard of living, the Laspeyres index is an upper bound on the true index and the Paasche index a lower bound [5]. The reason is intuitive once seen: a base-period basket over-weights precisely the goods that later became expensive, because the household bought plenty of them when they were cheap and has since bought fewer. Pricing that stale basket charges the household for a pattern of consumption it has already abandoned.

BEFORE READING ON

Two goods, priced at one apiece, with a household spending equally on both. One price doubles; the other does not move. The household is compensated just enough to stay exactly as well off as before. What does a Laspeyres index report, what does a Paasche index report, and what is the true answer?

If preferences are such that the household keeps spending half its money on each good — the unit-elasticity case — then the arithmetic is unusually clean. Laspeyres holds quantities at their old levels and reports fifty per cent. Paasche uses the new, substituted-away quantities and reports 33.3 per cent. The true index is $\sqrt{2}$, or 41.42 per cent. And the Fisher index, defined as the geometric mean of the other two, is $\sqrt{1.5 \times 1.33} = \sqrt{2}$ — exactly right, not approximately.

That is not a coincidence, and it is the central result in the theory of index numbers. W. Erwin Diewert showed in 1976 that a small family of formulas — Fisher's, Törnqvist's, Walsh's — are *superlative*: they give a second-order approximation to the true cost-of-living index for a broad class of preferences, without the statistician needing to know what those preferences are [6]. Making equal use of the quantities from both periods, rather than freezing one, is most of what buys that accuracy.

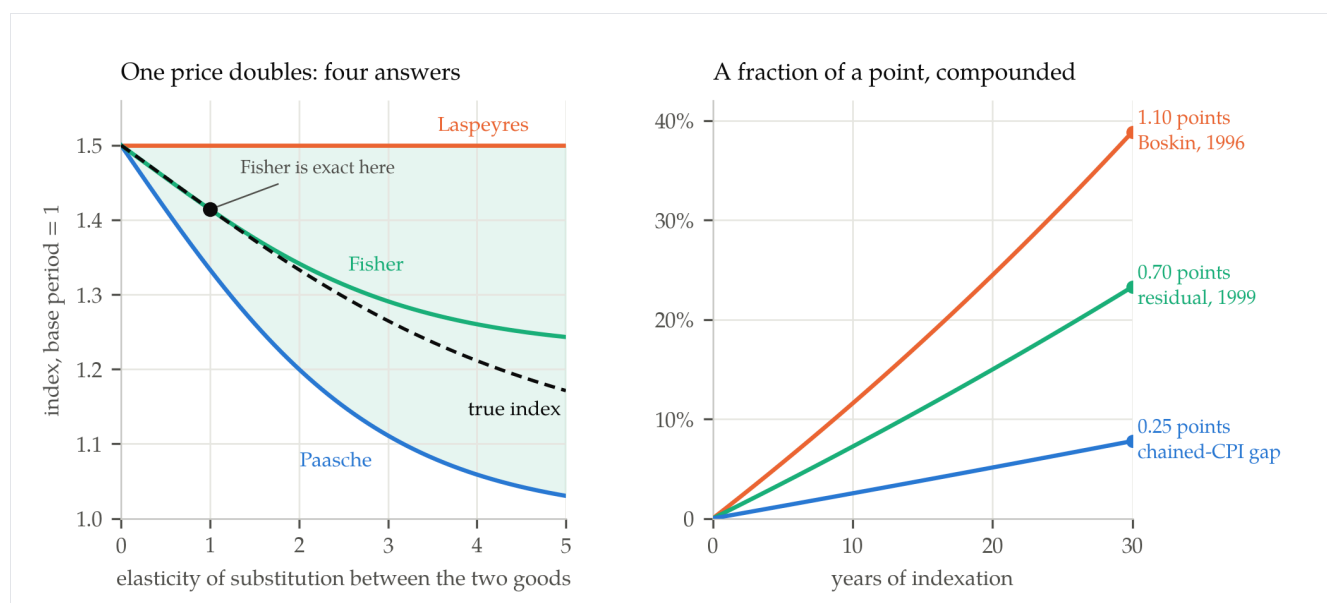


FIGURE Left: a symmetric two-good economy in which one price doubles, with the four indexes plotted against the elasticity of substitution. Laspeyres never moves, because the basket never moves. Right: the same fractional gap in an annual index, compounded over thirty years of indexation.

Two caveats survive all of this. Superlative is not exact: as the figure's left panel shows, once substitution becomes extreme the Fisher index drifts away from the truth it brackets. And K depends on the reference utility u , so there is a family of true indexes rather than one — the cost of last year's living standard and the cost of this year's are different numbers. The Bureau is careful to describe its target as a *conditional* cost-of-living index, holding constant everything that affects welfare but is not bought in a market: air quality, public goods, leisure, the weather [7].

Eighty Thousand Prices, Every Month

The machinery built on top of that theory is enormous and surprisingly physical. The American index is assembled from 243 item strata crossed with 32 geographic index areas, which yields 7,776 basic indexes to be estimated separately every month [8]. Roughly eighty thousand goods and services are priced in seventy-five urban areas, drawn from about six thousand housing units and twenty-three thousand retail establishments; the annual total runs to about a million individual prices [8], [9]. The CPI-U population covers about ninety-three per cent of the United States. Expenditure weights come from the Consumer Expenditure Survey, and since the January 2023 index they have been refreshed every year from a single calendar year of spending, replacing a biennial update built on two years of data — a change the Bureau estimated would remove roughly thirteen per cent of the remaining upper-level substitution bias and lower the twelve-month change by 0.036 percentage points [10].

That last figure is a useful calibration. The Bureau reports a median standard error of 0.07 per cent on the twelve-month change in the all-items index, so the sampling uncertainty around a published inflation rate is about a seventh of a point either way [9]. Almost every quantity argued over in the

literature on index bias is larger than that. The CPI's problem has never been that it counts badly; its problem is that what to count is a question with no purely technical answer.

The formula choice reaches all the way down. Below the level where expenditure weights exist, the index must average the price changes of individual items within a stratum — different cuts of beef, different shirts in different shops — and here the Bureau made an explicit substitution assumption. From January 1999 it replaced the arithmetic average of price relatives with a geometric average for the strata representing about sixty-one per cent of the index's weight, keeping the arithmetic form for shelter (27.3 per cent of weight), utilities and government charges (7.1 per cent) and medical care services (4.2 per cent), where substitution is genuinely limited [11]. The two formulas are not neutral alternatives: the arithmetic mean of relatives implies an elasticity of substitution of zero, the geometric mean implies exactly one. The change was expected to slow the annual rate of increase by roughly 0.2 percentage points [11], and the Bureau's later accounting puts the realised effect at somewhat under 0.3 [12]. Choosing a formula, stratum by stratum, is choosing how substitutable a country's beef cuts are.

The Hardest Word Is “Same”

Everything so far assumes the statistician can find the same good next month. That assumption is where price measurement genuinely breaks, and the best demonstration of how badly is William Nordhaus's history of artificial light. Nordhaus tracked what it actually cost to produce a given quantity of illumination, from the open fire through tallow candles, town gas, kerosene, Edison's carbon filament and the compact fluorescent tube. Measured in the wage-earner's own currency, the labour price of a thousand lumen-hours fell from 5.37 hours of work in 1800 to 0.000119 hours in 1992 — a factor of some forty-five thousand. A conventional index, tracking candles and then light bulbs as separate unrelated commodities, overstates the true price increase of light across that period by a factor of between nine hundred and sixteen hundred: an error of about 3.6 to 3.9 per cent a year, compounded for two centuries [13].

New goods are the same problem in sharper form, because before a good exists its price is not high — it is undefined. Jerry Hausman's proposal was to estimate the *virtual* price at which demand for the new item would have been exactly zero, and treat the fall from there to the launch price as a price decline the index should record. Applying the method to breakfast cereal, using the 1989 introduction of Apple-Cinnamon Cheerios as the test case, he concluded that the cereal index overstates inflation by roughly twenty-five per cent, or twenty once the imperfect competition of the cereal aisle is allowed for [14].

Both problems were correctly diagnosed long ago. The Price Statistics Review Committee chaired by George Stigler reported in 1961 that the index should abandon the fixed-basket framework for a constant-utility one, and that quality change should be handled by regression on product characteristics — hedonic methods, decades before the Bureau had the data or the computing to run them [1].

A DIRECTION WORTH CHECKING

Hedonic quality adjustment has a public reputation as the technique that quietly suppresses measured inflation: strip out the improvements, and the official price never seems to rise. In the American index, which way does hedonic adjustment actually push the headline number?

Upward, and barely at all. The Bureau's own accounting is that hedonic models outside the shelter and apparel components — that is, the ones applied to personal computers, televisions and similar durables since 1998 — cover about one per cent of the index's weight and have raised the annual rate of change of the all-items index by around 0.005 percentage points. In shelter and apparel, where hedonic adjustment has been used for decades, it usually *increases* the measured rate of change rather than lowering it [12]. The real quality-adjustment problem is not that the technique is being abused. It is that it is applied to almost nothing, because for most of the basket nobody knows how to write down the characteristics that would go into the regression.

One Point One

Which brings us back to the five economists. The Advisory Commission to Study the Consumer Price Index — Michael Boskin as chairman, with Ellen Dulberger, Robert Gordon, Zvi Griliches and Dale Jorgenson — reported to the Senate Finance Committee on 4 December 1996 that the index overstated the change in the cost of living by 1.1 percentage points a year, with a plausible range of 0.8 to 1.6 [2]. The decomposition is where the report's real content sits.

Source of bias	1996		Where it stood by 1999
	estimate	What was done about it	
Lower level substitution	0.25	Geometric mean formula, 1999	About 0.05 left
Upper level substitution	0.15	Chained index; more frequent weights	Largely addressed
New outlets	0.10	Nothing	Unchanged
Quality change, new products	0.60	More hedonic models	Still the largest, still contested

The commission also made an argument that had nothing to do with economics and everything to do with why anyone was listening. Left uncorrected, the report argued, over-indexation would grow into the fourth largest item in the federal budget, behind only Social Security, health care and defence, adding something like 1.07 trillion dollars to the national debt by 2008 [2]. The right-hand panel of the figure is that argument in one line: 1.1 points a year, compounded across thirty years of indexation, is a thirty-nine per cent divergence between what an indexed benefit pays and what it was meant to buy.

The Bureau made seven methodological changes over the following three years, and in 1999 the General Accounting Office asked the commissioners to re-estimate. Four of the five responded — Griliches was too ill — with remaining-bias figures between 0.73 and 0.90 percentage points [15]. All agreed the geometric-mean change had largely retired the lower-level problem and that quality change remained the largest and least tractable term. Governor Edward Gramlich of the Federal Reserve, testifying in April 1998, made the same triage explicit: fix the aggregation formulas first, because the bias there is quantifiable and the conceptual problems are not [16]. A decade on, Gordon revisited the report and judged it too conservative — the 1995-96 bias should have been put at 1.2 to 1.3 points, he argued, partly because the gap that opened between the traditional index and the chained one after 2000 was larger than anyone had forecast [17].

That chained index is the commission's most durable legacy. Introduced in August 2002 with a December 1999 base, the C-CPI-U applies a Törnqvist formula to expenditure weights from both periods, which is to say it is superlative by construction [18]. The Congressional Budget Office expects it to run about 0.25 percentage points a year below the traditional index [19]; the Bureau's own retrospective simulations put the gap at 0.32 points over 1990 to 2000 and 0.40 over 1996 to 2000 [18]. In the Tax Cuts and Jobs Act of 2017 Congress switched the indexation of the tax code to the chained measure, a change scored at roughly 134 billion dollars of additional revenue over ten years [20] — an appropriations decision executed entirely through a formula. It also demonstrated that substitution runs both ways: in 2020 the chained index rose *faster* than the traditional one, because the collapse in spending on air travel and similar services meant the categories gaining weight were the ones whose prices were not falling [20].

Whose Cost of Living

There is a final assumption buried under all of this, which is that a country has *a* cost of living. It does not. Using retail scanner data covering 2004 to 2015, Xavier Jaravel found that annual inflation for retail products ran 0.661 percentage points higher for households in the bottom income quintile than for those in the top, widening to about 0.88 points once changes in product variety are counted; across the broader consumption basket the gap is 0.368 points a year [21]. His explanation is a supply-side one: rising incomes at the top drew product innovation and competitive entry toward the goods high-income households buy, and that competition pushed their prices down.

Compounded over a working life, a third of a point is not a rounding error, and it lands on the people least able to absorb it. The mismatch shows up institutionally too. Social Security benefits are indexed not to the broad CPI-U but to the CPI-W, whose reference population is urban wage earners and clerical workers — about twenty-nine per cent of Americans, and not the twenty-nine per cent drawing retirement benefits [8]. That was a natural instrument for a wage-replacement programme when it was chosen, and it has simply stayed.

When the Index Is Captured

All of the above assumes a statistical agency trying to get the answer right. Argentina spent nearly nine years demonstrating what happens otherwise. After the government intervened at the national statistics institute in February 2007 and replaced the staff responsible for the price index, official inflation averaged about eight per cent a year over 2007 to 2011 while an index built from prices scraped from Argentine supermarkets' own websites showed over twenty. The manipulation was not even subtle: the online series tracked the *shape* of Argentine inflation faithfully, and one could approximate the official number by dividing the honest one by three. The International Monetary Fund censured Argentina in 2013; the practice ended with a change of administration in December 2015 [22].

The lesson is not that price indexes are unreliable, but that what makes one trustworthy is external replicability rather than internal sophistication. A superlative formula computed on numbers nobody outside the building can check is worth less than a crude Laspeyres index computed on data anyone can audit.

Meany in 1944 and Boskin in 1996 were complaining about the same thing from opposite ends, and both were right about it. A consumer price index is not a measurement of a natural quantity in the way a thermometer reading is. It is a definition, chosen from a family of defensible alternatives, and the choice moves money between generations, between income groups, and between citizens and their treasury. That is why it keeps getting litigated, and why the litigants are usually not economists.

The mature response is not to keep hunting for the one correct index, because the theory says plainly that there is a family of them and no fact of the matter selects a member. It is to publish several, state clearly which question each answers, and force the choice of which one indexes a pension out of the technical appendix and into the open, where it can be argued about by the people it pays or does not pay. The formula is a policy. It should be defended as one.

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